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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

PROJECT REPORT

Diabetes Onset Prediction

Using Machine Learning

An End-to-End Predictive Analytics / KDD Study
on the Pima Indians and Early Stage Diabetes Datasets

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Course: Predictive Analytics

Project: End-to-End Predictive Analytics Model (KDD process)

Datasets: Pima Indians Diabetes (768 records) and UCI Early Stage Diabetes Risk Prediction (520 records)

Submitted: October 2026

Disclaimer: *This is an academic project. The model described here is not a medical diagnostic system and has not been clinically validated.*

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Abstract

This project develops end-to-end predictive models that estimate whether a person is likely to have diabetes, using two related but independent public datasets. The primary dataset is the Pima Indians Diabetes Dataset (768 patients, eight clinical measurements); the secondary dataset is the UCI Early Stage Diabetes Risk Prediction Dataset (520 questionnaire records with age, gender and 14 Yes/No symptoms). Because their features differ, the datasets are **not merged**: each is analysed separately with the same KDD process and the same evaluation philosophy. Physiologically impossible zeros in Pima were treated as missing values, and 269 exact duplicate records (51.7 %) in the secondary dataset were removed before splitting to prevent leakage. Every learned preprocessing step (imputation, scaling, encoding, and SMOTE for Pima) was fitted on training data only. Five classifiers (Logistic Regression, K-Nearest Neighbors, Decision Tree, Random Forest and XGBoost) were compared with 5-fold cross-validation and hold-out test sets. Using the same recall-aware selection rule, **Random Forest** was chosen for both datasets: on Pima it achieved test accuracy 0.747, recall 0.741 and ROC-AUC 0.809 (cross-validated ROC-AUC 0.829); on the Early Stage dataset it achieved accuracy 0.902, recall 0.914 and ROC-AUC 0.964 (cross-validated ROC-AUC 0.977). SHAP identified **plasma glucose** as the dominant predictor for Pima, and **polyuria** and **polydipsia** for the Early Stage dataset. The symptom-based dataset proved easier to predict, but the populations and feature types differ, so the comparison shows how one method behaves on two kinds of data rather than which data source is better.

Chapter 1 — Introduction

1.1 Background

Diabetes mellitus is a chronic metabolic disease characterised by raised blood glucose, caused by insufficient insulin production, reduced sensitivity to insulin, or both. The World Health Organization describes diabetes as a major cause of blindness, kidney failure, heart attacks, stroke and lower-limb amputation [1]. Type 2 diabetes, the most common form, often develops gradually and may remain undiagnosed for years, so many people are identified only after complications have begun.

Health systems already collect simple measurements such as plasma glucose, body-mass index (BMI), blood pressure and age. Predictive analytics uses historical records of such measurements with known outcomes to learn patterns that can estimate risk for new individuals. In a screening setting, a model of this kind could help prioritise who should undergo confirmatory clinical testing. Questionnaires about typical symptoms are another low-cost source of information, used in the secondary dataset of this project.

1.2 Problem Statement

Given a vector of diagnostic measurements **X** = (Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age) for a patient, predict the binary outcome $\mathbf{y} \in \{0, 1\}$, where 1 indicates diabetes, and estimate the probability $P(y = 1 \mid X)$. This is a **supervised binary classification** problem on imbalanced, partially incomplete tabular data.

The same task is studied on a **second, independent dataset** (Section 1.7), in which the inputs are age, gender and 14 self-reported Yes/No symptoms, and the target is a Positive/Negative diabetes label. The two datasets have different features, so a separate model is built for each and only the results are compared.

1.3 Motivation

- **Early identification.** Earlier detection of high-risk individuals allows earlier confirmatory testing and lifestyle or medical intervention.
- **Low-cost inputs.** The features are routine measurements, so a screening-support model would not require expensive tests.
- **Academic value.** The dataset contains realistic data-quality problems: hidden missing values encoded as zeros, class imbalance and skewed features. This makes it well suited to demonstrating every stage of the KDD process.
- **Two complementary datasets.** Clinical measurements (Pima) and self-reported symptoms (Early Stage) are two different kinds of screening information; analysing both shows how much the available information changes what a model can achieve.

1.4 Objectives

1. Understand and clean both datasets, including invalid zero values (Pima) and duplicate records (Early Stage).
2. Explore relationships between the measurements and the outcome using visual EDA.
3. Build a leak-free preprocessing pipeline (imputation, scaling, SMOTE) fitted on training data only.
4. Train and compare Logistic Regression, KNN, Decision Tree, Random Forest and XGBoost.
5. Evaluate the models with Accuracy, Precision, Recall, F1, ROC-AUC, confusion matrices and ROC curves.
6. Select a final model with a justified, screening-oriented criterion rather than accuracy alone.
7. Interpret the model with feature importance and SHAP, and discuss limitations.
8. Repeat the analysis on the independent secondary dataset and compare the results of the two datasets.

1.5 Scope

The project covers the full modelling cycle, from problem definition to interpretation, on two public datasets that are analysed independently. It produces academic screening models and a demonstration prediction function. It does **not** merge the datasets, and it does not include clinical validation, external validation on other populations, deployment, or exhaustive hyperparameter optimisation.

1.6 Primary Dataset: Pima Indians Diabetes Dataset

The Pima Indians Diabetes Dataset was originally collected by the US National Institute of Diabetes and Digestive and Kidney Diseases and first used in machine-learning research by Smith et al. [2]. It was historically distributed through the UCI Machine Learning Repository and is now widely available via Kaggle [3], [4]. The copy used here comes from the public repository of J. Brownlee [5]. All 768 subjects are females of Pima Indian heritage aged at least 21.

Feature	Description	Unit
Pregnancies	Number of times pregnant	count
Glucose	Plasma glucose concentration, 2 h in an oral glucose tolerance test	mg/dL
BloodPressure	Diastolic blood pressure	mm Hg
SkinThickness	Triceps skin-fold thickness	mm
Insulin	2-hour serum insulin	µU/mL
BMI	Body-mass index	kg/m²
DiabetesPedigreeFunction	Score summarising diabetes history in relatives	–
Age	Age	years

Feature	Description	Unit
Outcome	**0 = no diabetes, 1 = diabetes**	class

The dataset has 768 rows, 9 numeric columns, no NaN values, no duplicate rows, and a class distribution of **500 (65.1 %) non-diabetic and 268 (34.9 %) diabetic** patients.

1.7 Secondary Dataset: Early Stage Diabetes Risk Prediction Dataset

The secondary dataset was published in the UCI Machine Learning Repository in 2020 (DOI 10.24432/C5VG8H, licensed CC BY 4.0) [21] and introduced by Islam et al. [19]. It contains **520 records** collected through direct questionnaires from patients of Sylhet Diabetes Hospital, Bangladesh, and approved by a doctor.

Variable(s)	Description	Type
Age	Age of the person	numeric (16–90 years)
Gender	Male / Female	categorical
Polyuria, Polydipsia, sudden weight loss, weakness, Polyphagia, Genital thrush, visual blurring, Itching, Irritability, delayed healing, partial paresis, muscle stiffness, Alopecia, Obesity	14 self-reported symptoms	categorical (Yes / No)
class (target)	Positive = diabetes (1), Negative = no diabetes (0)	categorical

How it differs from Pima.

- **Type of information:** self-reported symptoms and demographics instead of laboratory measurements.
- **Population:** patients of one diabetes hospital in Bangladesh (both sexes, ages 16–90) instead of Pima Indian women in the USA.
- **Class balance:** 61.5 % of the records are positive (68.9 % after removing duplicates), so diabetes is the *majority* class, the opposite of Pima.
- **Data quality:** no missing values, but 269 of the 520 rows (51.7 %) are exact duplicates.

Why it was included. The faculty required one additional dataset. A symptom questionnaire is a different, low-cost screening source, so the secondary dataset provides an *independent comparison perspective*: the same modelling approach is applied to it and the behaviour of the models is compared. Because the variables are completely different, the datasets are **analysed independently and never merged**. Section 3.12 describes how it was processed and Sections 4.11–4.14 report its results.

Chapter 2 — Literature Review

The Pima dataset is one of the most widely used benchmarks for diabetes prediction. This chapter summarises representative published studies. The figures quoted below are the findings reported by those authors, obtained under their own preprocessing, splits and validation schemes. They are not results of this project and are not directly comparable to them. Sections 2.8 and 2.9 cover studies on the secondary (Early Stage) dataset.

2.1 Smith et al. (1988): ADAP neural network [2]

- **Problem:** Forecast the onset of diabetes mellitus within five years in a high-risk population.
- **Dataset:** The original Pima Indian data from the NIDDK (the source of the dataset used here). As described in the dataset documentation, 576 cases were used for training and 192 for testing.
- **Method:** ADAP, an early adaptive-learning neural-network-like algorithm, compared with logistic regression and a linear perceptron.
- **Main finding:** Sensitivity and specificity crossed at about **0.76**.
- **Relevance:** This study introduced the dataset and established that the measurements carry predictive signal. It also evaluated performance with sensitivity, specificity and ROC analysis rather than accuracy alone, the same reasoning applied in this project.

2.2 Maniruzzaman et al. (2017): Gaussian process classification [6]

- **Problem:** Compare machine-learning paradigms for classifying diabetes data.
- **Dataset:** Pima Indian diabetes dataset (768 subjects: 268 diabetic, 500 controls).
- **Method:** Gaussian-process (GP) classification with linear, polynomial and radial-basis kernels, compared with linear discriminant analysis, quadratic discriminant analysis and naïve Bayes. Evaluation used accuracy, sensitivity, specificity, PPV, NPV and ROC curves.
- **Main finding:** The GP-based model performed best. Its reported accuracy, sensitivity and specificity were about 82 %, 92 % and 63 %.
- **Relevance:** This study shows the trade-off between sensitivity and specificity on this dataset and the value of reporting several metrics, both central themes of this project.

2.3 Sisodia and Sisodia (2018): Comparison of classification algorithms [7]

- **Problem:** Predict diabetes at an early stage with standard classification algorithms.
- **Dataset:** Pima Indians Diabetes Dataset.

- **Method:** Decision Tree, Support Vector Machine and Naïve Bayes, evaluated with precision, recall, F-measure and accuracy.
- **Main finding:** Naïve Bayes achieved the highest reported accuracy, about 76.3 %.
- **Relevance:** This is a baseline comparison of simple classifiers on the same data. It motivates the comparison of several model families in this project and shows that a single decision tree is typically not the strongest model.

2.4 Zou et al. (2018): Decision trees, random forests and neural networks [8]

- **Problem:** Predict diabetes mellitus from routine physical-examination data.
- **Dataset:** Hospital physical-examination records from Luzhou, China (14 attributes). The Pima Indians dataset was used as a second dataset, reduced to 392 complete records after removing missing values.
- **Method:** Decision tree, random forest and neural network, with dimensionality reduction by PCA and mRMR feature selection, evaluated with five-fold cross-validation.
- **Main finding:** Random forest gave the highest accuracy: 0.8084 on the Luzhou data with all attributes, and 0.7604 on the Pima data (0.7721 with mRMR). Glucose alone gave reasonable accuracy on the Luzhou data, but the authors concluded that glucose alone was not sufficient for the best result.
- **Relevance:** This supports including Random Forest as a strong candidate and shows that, although glucose is the key predictor, combining it with other measurements improves prediction, consistent with this project's findings. It also handled missing Pima values by deleting rows, whereas this project uses imputation to keep all 768 records.

2.5 Hasan et al. (2020): Ensembling of machine-learning classifiers [9]

- **Problem:** Build a robust diabetes-prediction framework.
- **Dataset:** Pima Indian Diabetes Dataset.
- **Method:** Outlier rejection, missing-value filling, standardisation and feature selection, followed by k-NN, decision trees, random forest, AdaBoost, naïve Bayes, XGBoost and an MLP. Models were combined with AUC-based weights and evaluated with K-fold cross-validation.
- **Main finding:** The proposed AUC-weighted ensemble was the best-performing classifier, with a reported AUC of 0.950, sensitivity 0.789 and specificity 0.934.
- **Relevance:** This study demonstrates the importance of preprocessing (especially handling missing values) and of ensemble methods such as XGBoost, both used in this project. Its much higher AUC comes from a more elaborate pipeline (outlier removal,

feature selection, ensembling) and a different validation protocol, so it is not directly comparable.

2.6 Joshi and Dhakal (2021): Logistic regression and decision trees [10]

- **Problem:** Predict type 2 diabetes in Pima Indian women and identify its main predictors.
- **Dataset:** Pima Indians Diabetes Dataset.
- **Method:** Logistic regression for predictor selection, a decision tree, and cross-validation.
- **Main finding:** The preferred specification reached a prediction accuracy of 78.26 % (cross-validation error 21.74 %). Glucose, pregnancies, BMI, diabetes pedigree function and age were identified as significant predictors.
- **Relevance:** This study supports using Logistic Regression as an interpretable baseline, and its predictor list closely matches the most important features found in this project.

2.7 Kavakiotis et al. (2017): Review of machine learning in diabetes research [11]

- **Problem:** Survey how machine learning and data mining are applied in diabetes research.
- **Dataset:** A systematic review of many studies, not a single dataset.
- **Method:** Literature review covering prediction and diagnosis, complications, genetics, and healthcare management.
- **Main finding:** Prediction and diagnosis is the most common application, and supervised methods (especially support vector machines) dominate. Random forests and other ensembles are also widely used.
- **Relevance:** This review places the project within the broader research landscape and justifies the choice of supervised classifiers.

2.8 Islam et al. (2019): Early-stage diabetes risk prediction [19]

- **Problem:** Predict the likelihood of diabetes at an early stage using data-mining techniques.
- **Dataset:** 520 questionnaire records (age, gender and 14 symptoms) from patients of Sylhet Diabetes Hospital, Bangladesh. This is the dataset used here as the secondary dataset, later published in the UCI repository [21].
- **Method:** Data-mining classification algorithms applied to the symptom data.

- **Main finding:** The study introduced the dataset and showed that symptoms can be used to estimate diabetes risk. Its detailed accuracy figures are not reproduced here because they could not be verified from an accessible copy of the paper.
- **Relevance:** It is the original source of the secondary dataset and motivates using symptom data for screening.

2.9 Vakil et al. (2021): Explainable models for early-stage diabetes [20]

- **Problem:** Compare machine-learning methods for early-stage diabetes prediction and explain their predictions.
- **Dataset:** The Sylhet hospital dataset (the secondary dataset of this project).
- **Method:** Random Forest, Decision Tree, artificial neural networks, KNN, SVM and XGBoost, with SHAP for explanation.
- **Main finding:** Random Forest performed best, with a reported accuracy of 99 % (arXiv preprint).
- **Relevance:** It supports Random Forest and SHAP for this dataset. Because the original file contains many duplicate records, very high accuracies should be interpreted with care unless duplicates are handled; Section 4.14 shows how much duplicates inflate the results in this project.

2.10 Methodological foundations

- **KDD process.** Fayyad, Piatetsky-Shapiro and Smyth [12] defined KDD as the overall process of discovering useful knowledge from data: selection, preprocessing, transformation, data mining and interpretation. This project is structured around those stages.
- **SMOTE.** Chawla et al. [13] proposed generating synthetic minority-class samples by interpolating between neighbouring minority examples. This improves classifier sensitivity to the minority class compared with plain oversampling.
- **Random Forest.** Breiman [14] introduced random forests: ensembles of decision trees trained on bootstrap samples with random feature selection at each split. They reduce variance compared with a single tree.
- **XGBoost.** Chen and Guestrin [15] described a scalable, regularised gradient-tree-boosting system that is widely used for tabular data.
- **SHAP.** Lundberg and Lee [16] unified several explanation methods under Shapley values from cooperative game theory. This gives additive per-feature contributions to each prediction.
- **Software.** Scikit-learn [17] and imbalanced-learn [18] provide the implementations of pipelines, models and SMOTE used here.

2.11 Research gap addressed by this project

Published accuracies on the Pima dataset vary widely (roughly 76 % to above 90 %), partly because studies differ in how they treat the invalid zeros, whether they apply oversampling or scaling before splitting the data, and how they validate. This project does not aim to beat published numbers. Instead, it aims to produce a transparent, leak-free and reproducible baseline, in which every learned transformation (imputation, scaling, SMOTE) is fitted only on training data, and model selection explicitly considers recall. The same concern applies to the secondary dataset, where half of the records are exact duplicates. This project therefore builds leak-free pipelines for both datasets and compares them with one consistent evaluation method.

Chapter 3 — Workflow / Predictive Modelling Process

The project follows the KDD process (Figure 3.1), applied separately to each dataset. Sections 3.1–3.11 describe the primary (Pima) dataset; Section 3.12 describes the secondary (Early Stage) dataset. Each stage corresponds to a section of the notebook.

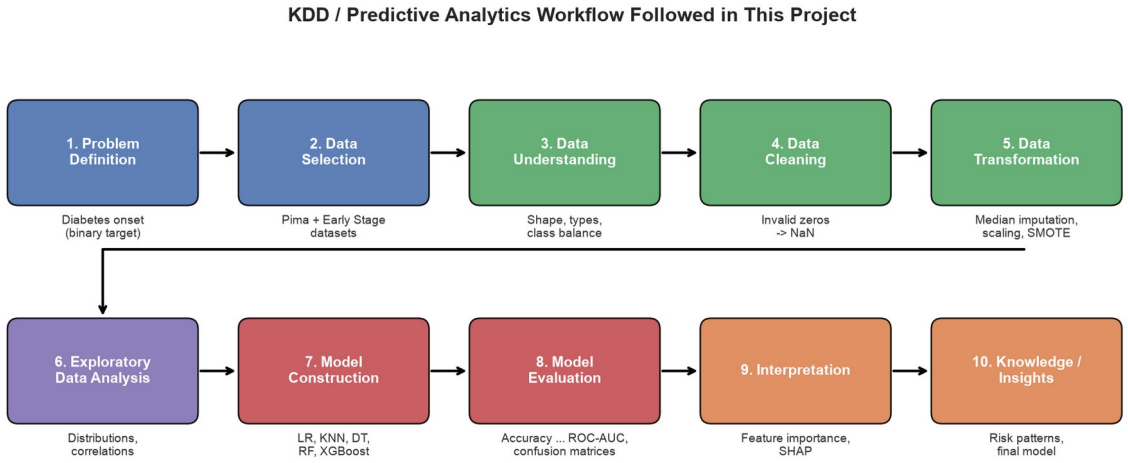


Figure 3.1: KDD / predictive analytics workflow followed in this project.

Problem Definition → Data Selection → Data Understanding → Data Cleaning → Data Transformation → Exploratory Data Analysis → Model Construction → Model Evaluation → Interpretation → Knowledge / Insights

3.1 Problem Definition

The task was framed as binary classification (Outcome 0/1) with a probability output. Because the intended use is *screening support*, missing a diabetic patient (false negative) was treated as more costly than a false alarm. Recall was therefore given special weight alongside the threshold-independent ROC-AUC.

3.2 Data Collection

The Pima Indians Diabetes Dataset was downloaded from the public repository [5]. The original file has no header, so the nine column names were added. The 768 observations were not altered, and all cleaning is performed in code. The notebook resolves the data path automatically and can re-download the file if it is missing. The secondary dataset is downloaded automatically from the UCI repository [21] by the notebook and saved to the data/ folder.

3.3 Data Understanding

head(), tail(), info(), describe(), dtypes, missing-value, duplicate and unique-value checks were performed. All features are numeric and there are no NaNs or duplicate rows. However, describe() revealed **minimum values of 0** in Glucose, BloodPressure, SkinThickness, Insulin and BMI, which is physiologically impossible. The target is imbalanced (65.1 % / 34.9 %).

3.4 Data Cleaning

Zero values in the five affected columns were replaced with NaN. Zeros in Pregnancies (a valid count) and Outcome (the negative class) were kept.

Feature	Invalid zeros	% of rows
Glucose	5	0.65
BloodPressure	35	4.56
SkinThickness	227	29.56
Insulin	374	48.70
BMI	11	1.43

Replacing 0 with NaN is a fixed rule that learns nothing from the data, so applying it before splitting cannot leak information. Rows were not dropped, because removing every incomplete row would discard roughly half the dataset.

3.5 Exploratory Data Analysis

EDA included the class count plot, histograms of all features, by-outcome density plots for Glucose, BMI and Age, boxplots of all features by outcome, a correlation heatmap and a pairplot of the four features most correlated with the outcome. Every plot has a title, axis labels and a legend where relevant. Findings are reported in Chapter 4.

3.6 Data Preprocessing

1. **Stratified split first.** train_test_split(X, y, test_size=0.2, stratify=y, random_state=42) produced 614 training rows (400 / 214) and 154 test rows (100 / 54). Both sets keep the original class ratio of about 35 % positives.
2. **Median imputation**, with medians learned from the training set (e.g. Insulin 125.0, SkinThickness 29.0, Glucose 117.0). The median was chosen because Insulin and SkinThickness are strongly right-skewed.
3. **Standardisation** (StandardScaler), with means and standard deviations learned from the training set.

The test set never influences any learned parameter. Otherwise, information about "future" patients would leak into training and inflate the evaluation scores.

3.7 Class Imbalance

SMOTE [13] was used to oversample the diabetic class **in the training data only**, from 400 / 214 to 400 / 400. The test set keeps its real 100 / 54 distribution. SMOTE was placed *inside* an imblearn pipeline after imputation and scaling, so during cross-validation it is re-applied to each training fold and never touches the validation fold.

class_weight="balanced" was compared as an alternative, using Logistic Regression and 5-fold CV on the training data:

Strategy	CV Accuracy	CV Recall	CV F1	CV ROC-AUC
No imbalance handling	0.7882	0.5748	0.6545	0.8434
class_weight="balanced"	0.7622	0.7010	0.6724	0.8444
SMOTE	0.7720	0.7244	0.6885	0.8427

Without imbalance handling, recall falls to 0.57, meaning that more than 4 in 10 diabetic patients would be missed. Both techniques raise recall by about 13–15 points at a small cost in accuracy, while ROC-AUC (a ranking measure) is essentially unchanged. SMOTE was adopted for all models because it applies uniformly, including to KNN, which has no class-weight option. Class weights were not combined with SMOTE, to avoid correcting the imbalance twice.

3.8 Model Selection

Five algorithms representing different learning paradigms were chosen:

Model	Paradigm	Configuration
Logistic Regression	Linear, probabilistic baseline	max_iter=1000
K-Nearest Neighbors	Instance-based (distance)	n_neighbors=15
Decision Tree	Rule-based tree	max_depth=5, min_samples_leaf=5
Random Forest	Bagging ensemble [14]	n_estimators=300, min_samples_leaf=2
XGBoost	Gradient-boosting ensemble [15]	n_estimators=200, max_depth=3, learning_rate=0.05, subsample=0.8, colsample_bytree=0.8

The settings are conservative defaults chosen to limit overfitting on a small dataset. No hyperparameters were tuned on the test set.

3.9 Model Training

Every model uses the same pipeline:

```
SimpleImputer(median) → StandardScaler → SMOTE → Classifier
```

Scaling is required for Logistic Regression and KNN. It is harmless for the tree models, because it does not change the possible splits, and it gives SMOTE comparable feature scales for its nearest-neighbour search. Each pipeline was first assessed with **5-fold stratified cross-validation on the training set**, then refitted on all 614 training rows. `random_state=42` was used throughout.

3.10 Evaluation

The fitted pipelines were evaluated once on the 154-patient test set using Accuracy, Precision, Recall, F1, ROC-AUC and Specificity, together with classification reports, confusion matrices and ROC curves. The final model was chosen with a predefined rule:

1. **Screening constraint:** recall ≥ 0.70 in both cross-validation and the test set.

2. **Overall score:** the mean of the five metrics, averaged equally over cross-validation and test. This rewards consistency rather than performance on one split.

3.11 Interpretation

The final model was interpreted with (a) impurity-based feature importance from the three tree models and the absolute standardised coefficients of Logistic Regression, and (b) SHAP values computed with TreeExplainer [16] on the test set. A permutation-importance fallback was implemented in case SHAP failed, but it was not needed.

3.12 Workflow for the Secondary Dataset

The secondary dataset follows the same stages, with steps adapted to its data:

- **Data collection:** the UCI zip file is downloaded automatically (with a copy in the project repository as a fallback).
- **Data understanding:** the columns are inspected before any preprocessing code is written, and the **target column is identified automatically** (class). Only Age is numeric; the other 15 features are text categories.
- **Data cleaning:** fixed text rules trim spaces, unify Yes/No spelling and turn "unknown"-style tokens into missing values (none were found). **269 exact duplicate rows (51.7 %)** were found, with no conflicting labels. They were removed **before** the split, leaving 251 unique records (173 positive, 78 negative). Otherwise copies of the same record would appear in both the training and the test set.
- **Preprocessing:** the target is encoded (Positive = 1). A stratified 80/20 split gives 200 training and 51 test records. Inside each model's pipeline a ColumnTransformer, fitted on training data only, applies median imputation and scaling to Age, and most-frequent imputation and one-hot encoding to the categories (binary columns become one 0/1 column).
- **Class imbalance:** SMOTE was not used, because it would create impossible values for 0/1 answers. Class weights were compared with no re-weighting by 5-fold cross-validation on the training data (table below). Because the minority class here is the *non-diabetic* class, balancing lowered recall, so no re-weighting was used.
- **Models:** the same five algorithms with the same settings as for Pima (Table in Section 3.8).
- **Evaluation and interpretation:** the same metrics, 5-fold cross-validation, the same recall-aware selection rule, impurity-based feature importance and SHAP.

ColumnTransformer (Age: median + scaling | categories: most-frequent + one-hot) → Classifier

Strategy (average of five models)	CV Accuracy	CV Precision	CV Recall	CV F1	CV ROC-AUC
No re-weighting (used)	0.8990	0.9473	0.9071	0.9240	0.9591
class_weight = "balanced"	0.8880	0.9581	0.8780	0.9140	0.9632

Chapter 4 — Results and Discussion

All values in this chapter come from the executed notebook and the files in the results/ folder. Sections 4.1–4.10 report the primary (Pima) dataset, Sections 4.11–4.13 the secondary (Early Stage) dataset, and Section 4.14 compares the two.

4.1 Dataset Statistics

Feature	Mean	Std	Min	Median	Max
Pregnancies	3.85	3.37	0	3.0	17
Glucose	120.89	31.97	0*	117.0	199
BloodPressure	69.11	19.36	0*	72.0	122
SkinThickness	20.54	15.95	0*	23.0	99
Insulin	79.80	115.24	0*	30.5	846
BMI	31.99	7.88	0*	32.0	67.1
DiabetesPedigreeFunction	0.47	0.33	0.078	0.37	2.42
Age	33.24	11.76	21	29.0	81

** Invalid zeros, treated as missing values. Statistics are computed on the raw data, before cleaning.*

4.2 Class Distribution

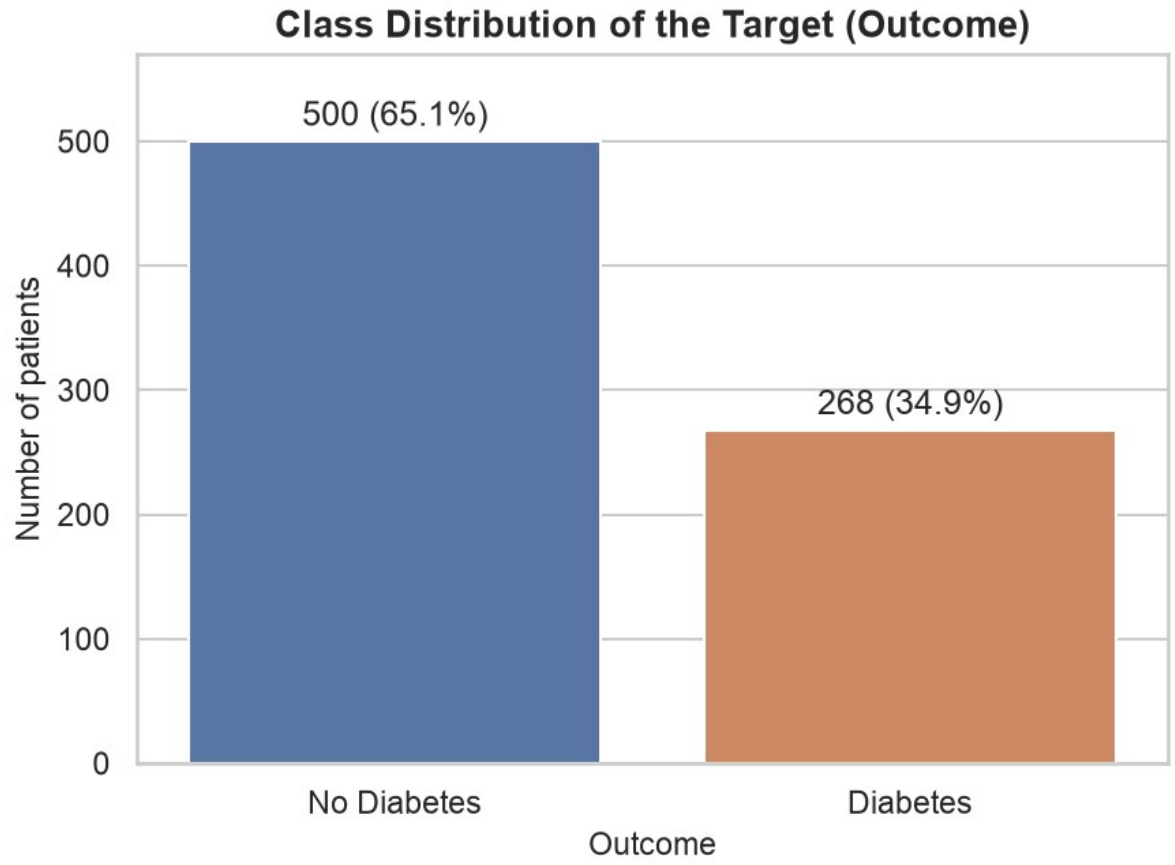


Figure 4.1: 500 non-diabetic (65.1 %) vs. 268 diabetic (34.9 %) patients.

A trivial classifier that always predicts "no diabetes" would score 65 % accuracy while detecting no diabetic patients. This is why accuracy is not used on its own.

4.3 EDA Findings

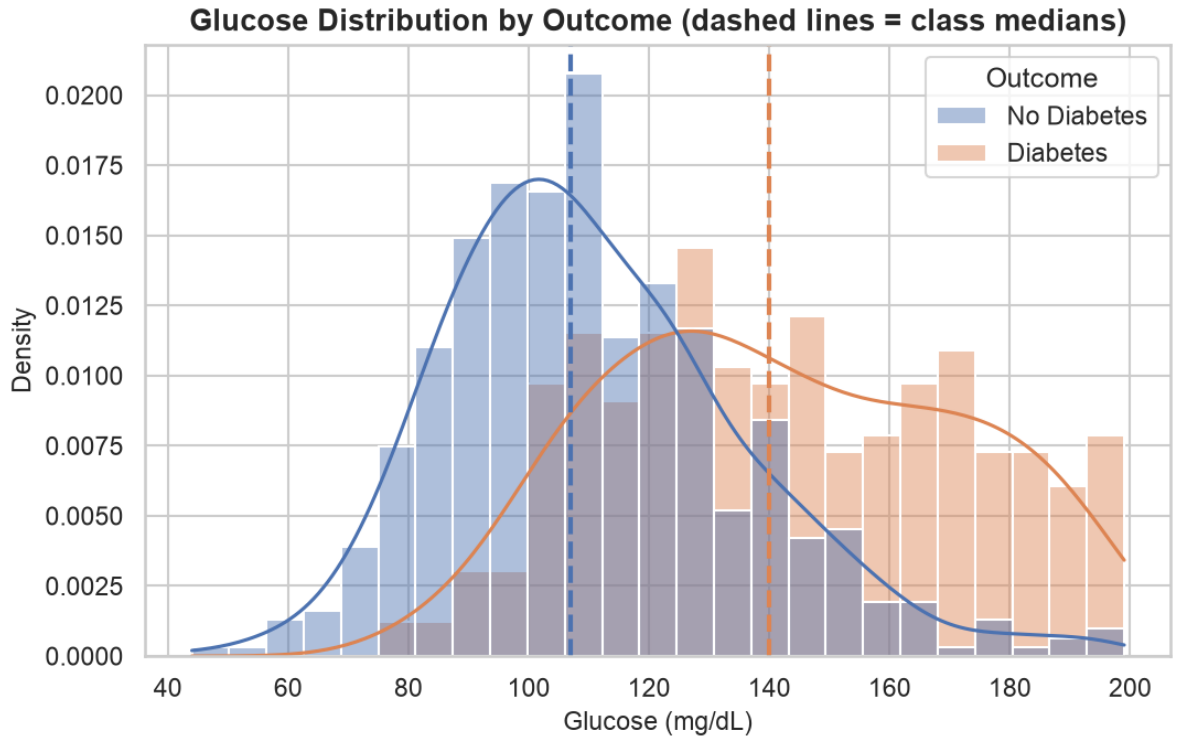


Figure 4.2: Glucose distribution by outcome (dashed lines = class medians).

Feature	Median (No Diabetes)	Median (Diabetes)	Correlation with Outcome
Glucose	107.0	140.0	0.495
BMI	30.1	34.3	0.314
Insulin	102.5	169.5	0.303
SkinThickness	27.0	32.0	0.259
Age	27.0	36.0	0.238
Pregnancies	2.0	4.0	0.222
DiabetesPedigreeFunction	0.34	0.45	0.174
BloodPressure	70.0	74.5	0.171

(Computed after replacing invalid zeros with NaN.)

- **Glucose** separates the classes most clearly (medians of 140 vs. 107 mg/dL; r = 0.495).
- **BMI** and **Age** are moderately higher in diabetic patients. Non-diabetic patients are concentrated in their 20s.
- **Insulin** and **SkinThickness** are higher in the diabetic group but have the most missing data.

- The strongest feature–feature correlations are SkinThickness–BMI (0.65), Glucose–Insulin (0.58) and Pregnancies–Age (0.54). These are moderate and do not justify removing features.
- Every feature shows substantial class overlap, so no single measurement is sufficient for classification.

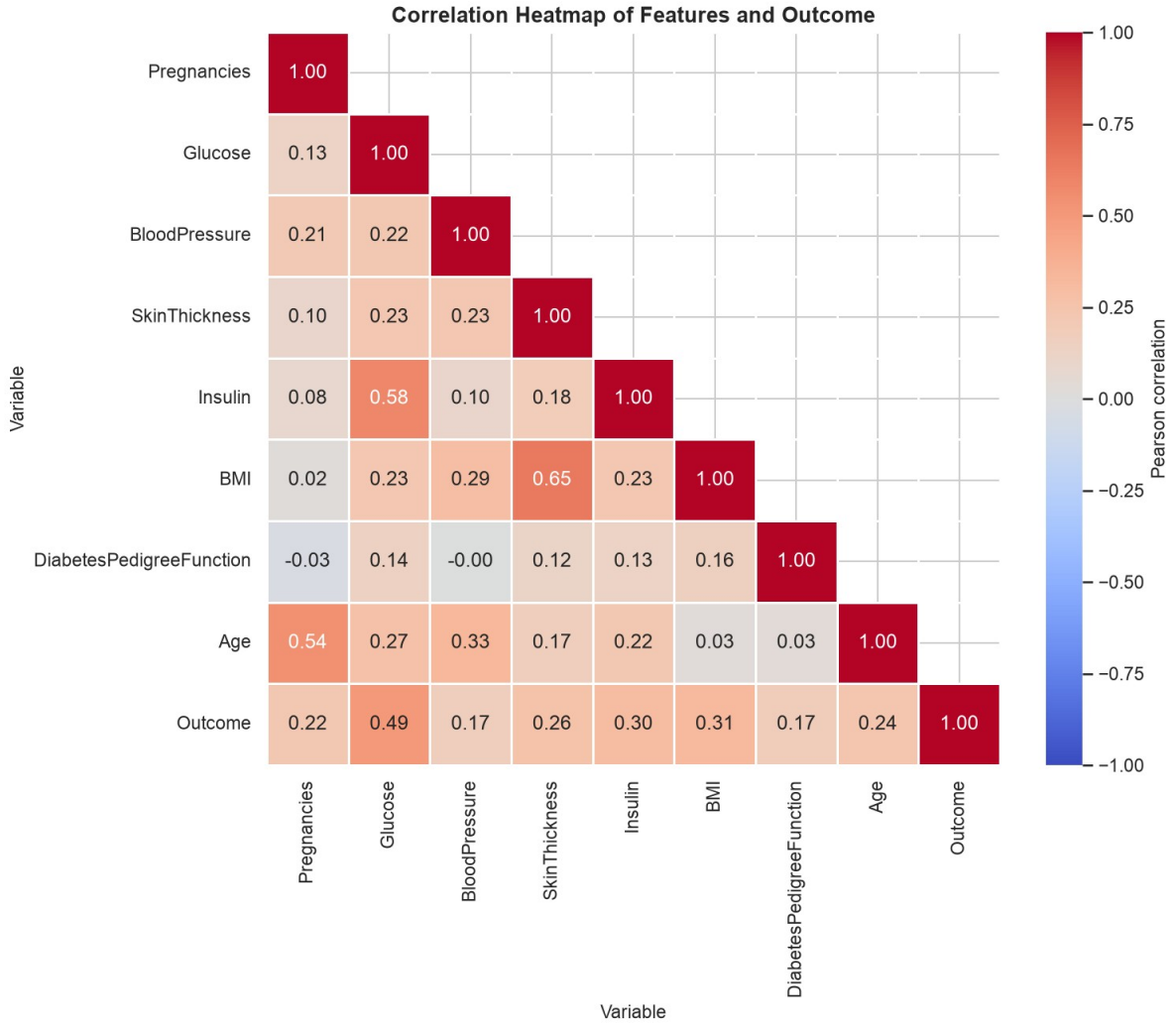


Figure 4.3: Correlation heatmap (pairwise-complete observations).

These relationships are **associations** in this sample, not evidence of causation.

4.4 Model Comparison

Cross-validation on the training set (5 folds, mean ± std):

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.772 ± 0.015	0.658 ± 0.026	0.724 ± 0.045	0.689 ± 0.021	0.843 ± 0.020
K-Nearest Neighbors	0.728 ± 0.054	0.588 ± 0.070	0.790 ± 0.020	0.672 ± 0.047	0.820 ± 0.045
Decision Tree	0.697 ± 0.029	0.560 ± 0.047	0.687 ± 0.069	0.612 ± 0.013	0.762 ± 0.013
Random Forest	0.779 ± 0.009	0.672 ± 0.022	0.715 ± 0.028	0.692 ± 0.009	0.829 ± 0.025

Model	Accuracy	Precision	Recall	F1	ROC-AUC
XGBoost	0.751 ± 0.027	0.632 ± 0.041	0.687 ± 0.058	0.657 ± 0.040	0.822 ± 0.030

Hold-out test set (154 patients), sorted by ROC-AUC (results/model_comparison.csv):

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Specificity
XGBoost	0.7532	0.6250	0.7407	0.6780	0.8263	0.760
Logistic Regression	0.7078	0.5738	0.6481	0.6087	0.8113	0.740
Random Forest	0.7468	0.6154	0.7407	0.6723	0.8094	0.750
K-Nearest Neighbors	0.6948	0.5455	0.7778	0.6412	0.7974	0.650
Decision Tree	0.7273	0.5833	0.7778	0.6667	0.7616	0.700

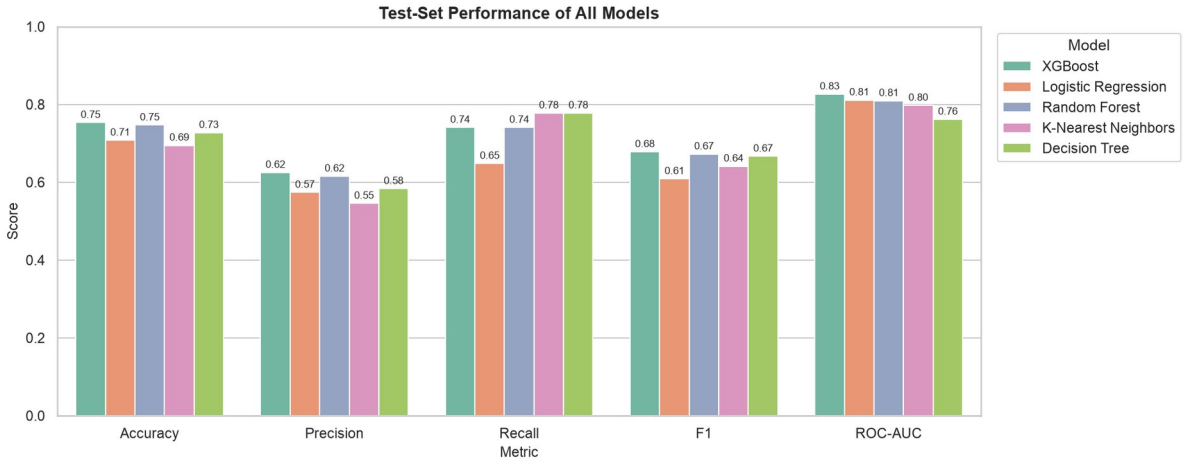


Figure 4.4: Test-set metrics for all five models.

Best model per metric (test set): Accuracy: XGBoost (0.7532). Precision: XGBoost (0.6250). Recall: KNN and Decision Tree (0.7778). F1: XGBoost (0.6780). ROC-AUC: XGBoost (0.8263).

4.5 Confusion Matrices

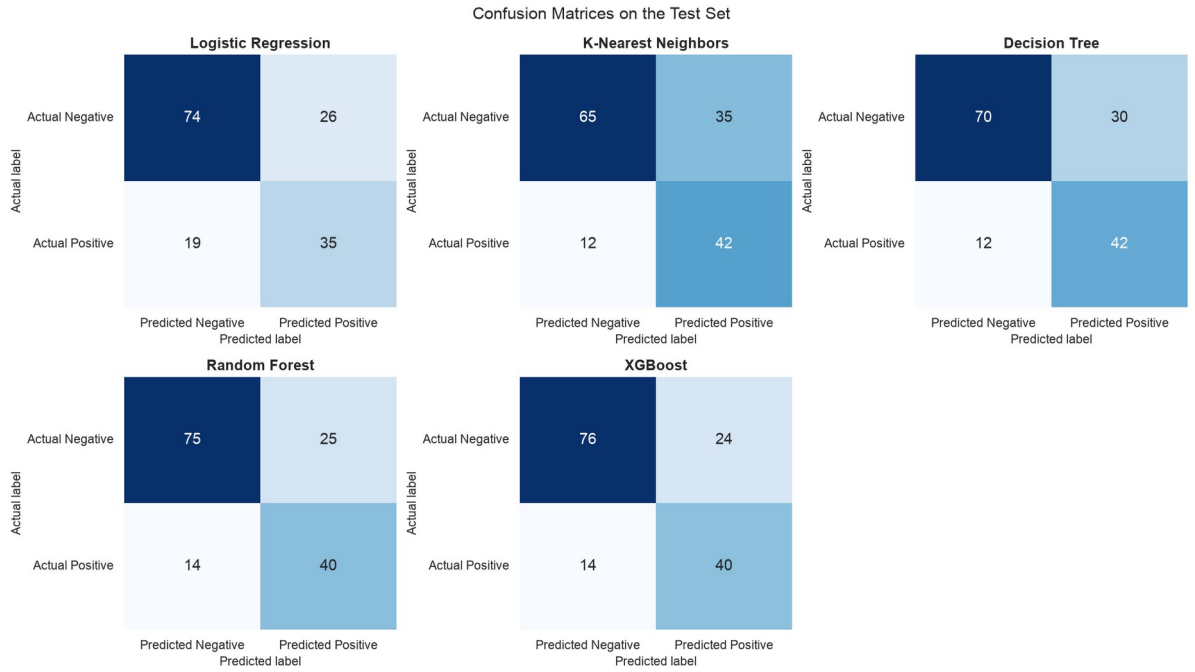


Figure 4.5: Confusion matrices on the test set (54 diabetic, 100 non-diabetic patients).

Model	TP	TN	FP	FN
XGBoost	40	76	24	14
Logistic Regression	35	74	26	19
Random Forest	40	75	25	14
K-Nearest Neighbors	42	65	35	12
Decision Tree	42	70	30	12

KNN and the Decision Tree miss the fewest diabetic patients (12 FN) but raise the most false alarms (35 and 30 FP). Logistic Regression misses the most (19 FN). Random Forest and XGBoost sit in between: 14 FN with 24–25 FP. In screening, a false negative may delay referral for confirmatory testing, while a false positive leads to an additional test that would likely reveal the error. The aim is therefore a low FN count without an unmanageable FP count.

4.6 ROC-AUC Comparison

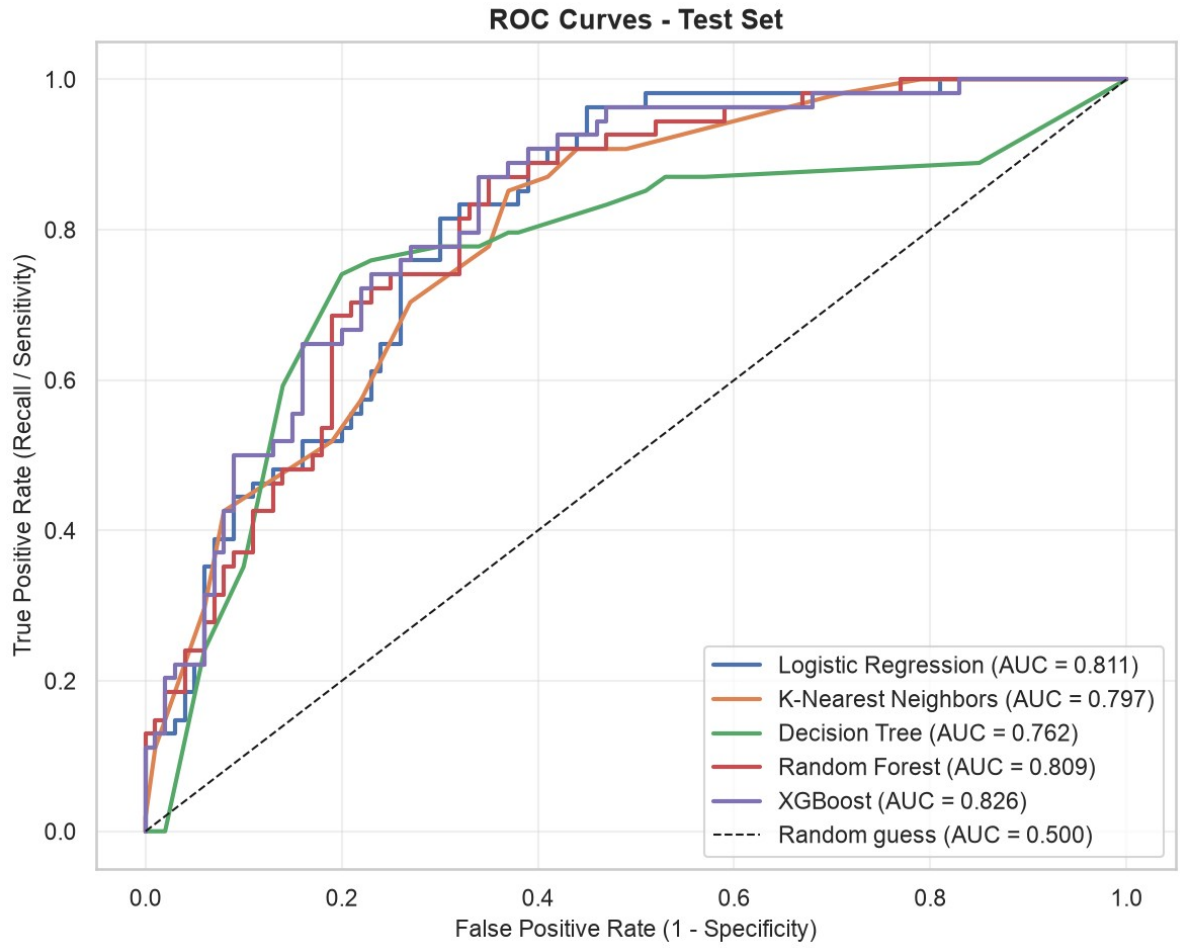


Figure 4.6: ROC curves of all models on the test set.

All models are well above the random baseline (AUC 0.5). XGBoost (0.826), Logistic Regression (0.811) and Random Forest (0.809) have nearly overlapping curves. The single Decision Tree is lowest (0.762), because a shallow tree produces only a few distinct probability values. ROC-AUC can be read as the probability that a randomly chosen diabetic patient receives a higher risk score than a randomly chosen non-diabetic patient.

4.7 Final Model Selection

Model	Test mean (5 metrics)	CV mean (5 metrics)	Test Recall	CV Recall	Overall score	Recall ≥ 0.70 in both
Random Forest	0.7169	0.7374	0.7407	0.7148	**0.7272**	**Yes**
XGBoost	0.7246	0.7096	0.7407	0.6866	0.7171	No
K-Nearest Neighbors	0.6913	0.7196	0.7778	0.7898	0.7054	Yes
Logistic Regression	0.6699	0.7372	0.6481	0.7244	0.7036	No
Decision Tree	0.7033	0.6637	0.7778	0.6870	0.6835	No

Random Forest was selected. It is the only model that ranks first or second on most metrics in *both* cross-validation and the test set, and it meets the recall constraint in both. XGBoost was best on four test metrics, but its cross-validated recall (0.687) was below the threshold. Logistic Regression had the best cross-validated ROC-AUC but the lowest test recall. KNN met the recall constraint but had clearly lower precision (0.545) and specificity (0.650).

Final model performance (test set): Accuracy 0.7468 · Precision 0.6154 · Recall 0.7407 · F1 0.6723 · ROC-AUC 0.8094 · Specificity 0.750. Cross-validated ROC-AUC: 0.829 ± 0.025.

4.8 Feature Importance

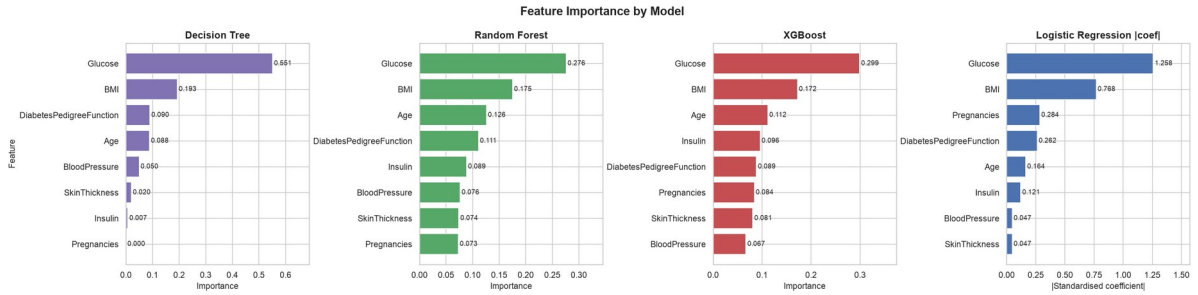


Figure 4.7: Feature importance of the tree models and |standardised coefficients| of Logistic Regression.

Feature	Decision Tree	Random Forest	XGBoost	LR |coef|
Glucose	0.551	0.276	0.299	1.258
BMI	0.193	0.175	0.172	0.768
Age	0.088	0.126	0.112	0.164
DiabetesPedigreeFunction	0.090	0.111	0.089	0.262
Insulin	0.007	0.089	0.096	0.121
BloodPressure	0.050	0.076	0.067	0.047
SkinThickness	0.020	0.074	0.081	0.047
Pregnancies	0.000	0.073	0.084	0.284

All models agree that **Glucose** is the most important feature, and **BMI** is consistently second. Age and DiabetesPedigreeFunction follow, while BloodPressure, SkinThickness and Pregnancies contribute least in the ensemble models. Impurity-based importance can favour continuous features and is shared between correlated features, so it shows how much the model *relies on* a feature, not causation.

4.9 SHAP Interpretation

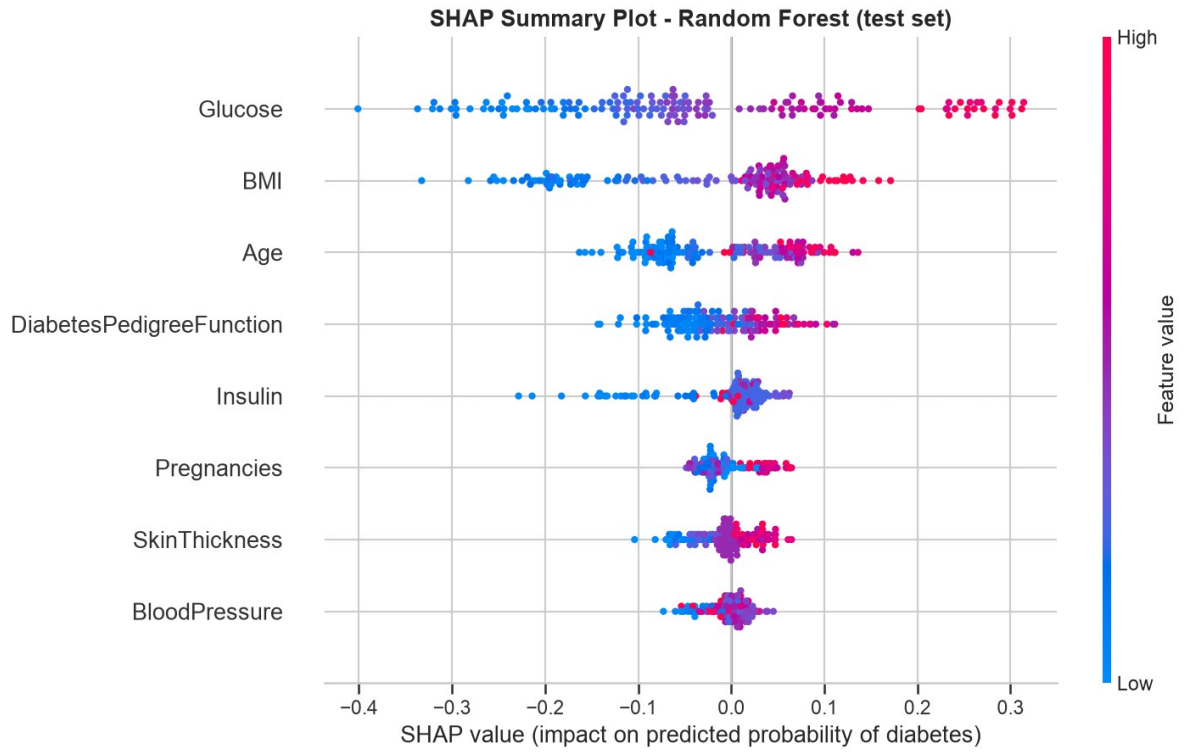


Figure 4.8: SHAP summary (beeswarm) plot for the Random Forest on the test set. Colour = feature value (red high, blue low).

Feature	Mean |SHAP|	Corr(feature value, SHAP)	Higher value pushes prediction
Glucose	0.146	0.955	toward Diabetes
BMI	0.094	0.837	toward Diabetes
Age	0.065	0.733	toward Diabetes
DiabetesPedigreeFunction	0.043	0.682	toward Diabetes
Insulin	0.034	0.237	toward Diabetes (weak)
Pregnancies	0.025	0.635	toward Diabetes
SkinThickness	0.024	0.778	toward Diabetes
BloodPressure	0.015	0.255	weak / mixed

- **Positive contributions.** High glucose has the largest positive SHAP values, raising the predicted probability by up to about 0.3. High BMI and older age also push predictions toward diabetes.
- **Negative contributions.** Low glucose, low BMI and younger age push predictions toward no diabetes. Low Insulin values produce clearly negative contributions, while high Insulin values have little effect.
- **Relative importance.** The SHAP ranking (Glucose > BMI > Age > DPF > others) matches the impurity-based ranking, and it is consistent with the EDA correlations and with the predictors reported by Joshi and Dhakal [10].

- SHAP explains how *this model* uses the features. It does not establish medical causality, and correlated features can share credit.

4.10 Sample Prediction

The predict_diabetes() function applies the same zero-to-NaN rule and the full fitted pipeline to one patient. Three illustrative (hypothetical) inputs gave:

Example	Key inputs	Predicted class	P(diabetes)
A	Glucose 168, BMI 38.2, Age 50	Diabetes likely	87.2 %
B	Glucose 89, BMI 24.5, Age 24	Diabetes unlikely	1.7 %
C	Glucose 130, BMI 32.0, Age 35, Insulin and SkinThickness not measured	Diabetes likely	66.7 %

These are academic demonstrations, not medical diagnoses. The probabilities are not calibrated.

4.11 Secondary Dataset: Data Understanding and EDA

The 520 raw records contain no missing values but 269 exact duplicates. After removing them, 251 unique records remain: **173 diabetic (68.9 %)** and **78 non-diabetic (31.1 %)**. The EDA uses the unique records, so that duplicated people are not counted twice.

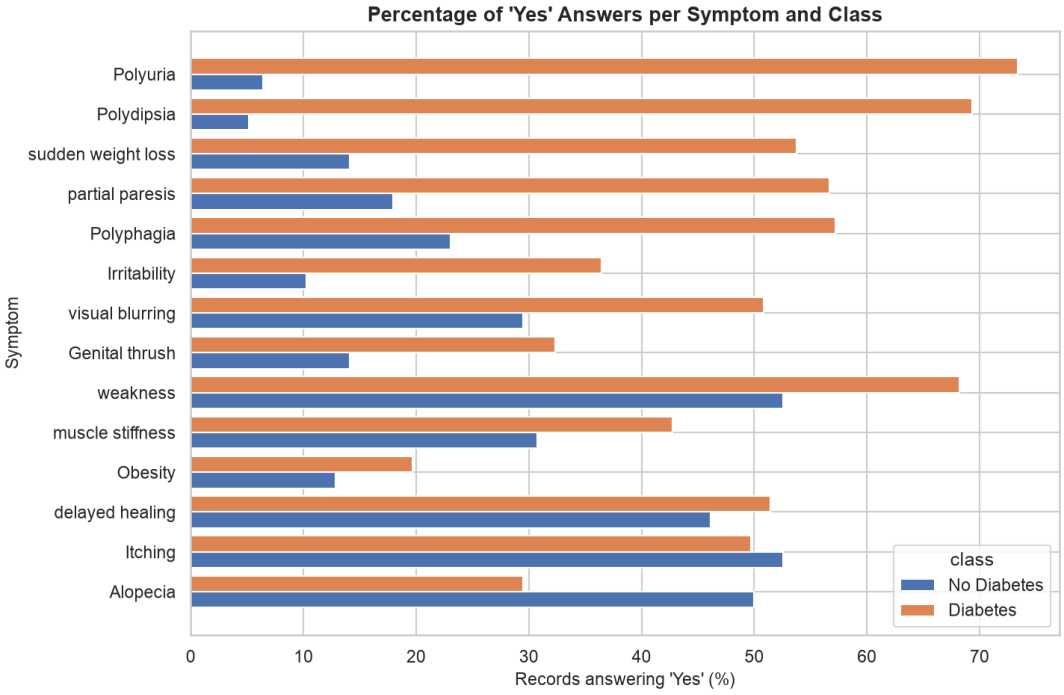


Figure 4.9: Percentage of "Yes" answers per symptom among diabetic and non-diabetic records (unique records).

Feature	"Yes" in non-diabetic (%)	"Yes" in diabetic (%)	Association with class
Polyuria	6.4	73.4	0.621

Feature	"Yes" in non-diabetic (%)	"Yes" in diabetic (%)	Association with class
Polydipsia	5.1	69.4	0.595
sudden weight loss	14.1	53.8	0.373
partial paresis	17.9	56.6	0.360
Polyphagia	23.1	57.2	0.317
Alopecia	50.0	29.5	−0.198
Gender (Male)	85.9	53.8	−0.309

- **Polyuria** (excessive urination) and **Polydipsia** (excessive thirst) separate the classes most strongly.
- 88 % of the women's records are diabetic against 58 % of the men's, a property of this hospital sample.
- Age differs little between the classes (median 49 vs 46.5 years), unlike in Pima.
- Several features separate the classes strongly, so the secondary dataset is expected to be easier to predict than Pima.

4.12 Secondary Dataset: Model Comparison

Cross-validation on the training set (200 records, 5 folds, mean ± std):

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.890 ± 0.025	0.928 ± 0.028	0.913 ± 0.059	0.919 ± 0.022	0.972 ± 0.009
K-Nearest Neighbors	0.860 ± 0.030	0.984 ± 0.020	0.811 ± 0.056	0.887 ± 0.031	0.966 ± 0.023
Decision Tree	0.885 ± 0.030	0.924 ± 0.048	0.913 ± 0.056	0.916 ± 0.024	0.900 ± 0.045
Random Forest	0.925 ± 0.027	0.951 ± 0.027	0.942 ± 0.049	0.945 ± 0.021	0.977 ± 0.013
XGBoost	0.935 ± 0.020	0.950 ± 0.016	0.957 ± 0.035	0.953 ± 0.015	0.980 ± 0.007

Hold-out test set (51 records: 35 diabetic, 16 non-diabetic), sorted by ROC-AUC (results/early_stage_model_comparison.csv):

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Specificity
Random Forest	**0.9020**	**0.9412**	**0.9143**	**0.9275**	**0.9643**	**0.875**
Decision Tree	0.8824	0.9677	0.8571	0.9091	0.9562	0.938
XGBoost	0.8824	1.0000	0.8286	0.9062	0.9464	1.000
K-Nearest Neighbors	0.7843	1.0000	0.6857	0.8136	0.9259	1.000
Logistic Regression	0.8235	0.9333	0.8000	0.8615	0.9250	0.875

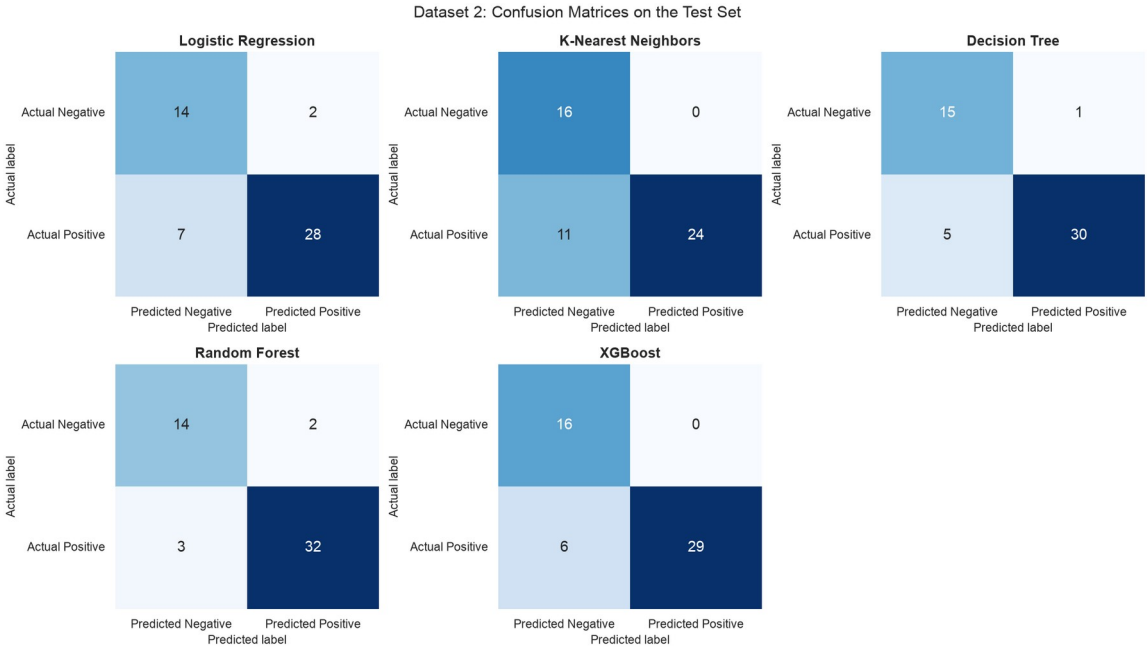


Figure 4.10: Confusion matrices of the secondary dataset on its test set.

Model	TP	TN	FP	FN
Random Forest	32	14	2	3
Decision Tree	30	15	1	5
XGBoost	29	16	0	6
K-Nearest Neighbors	24	16	0	11
Logistic Regression	28	14	2	7

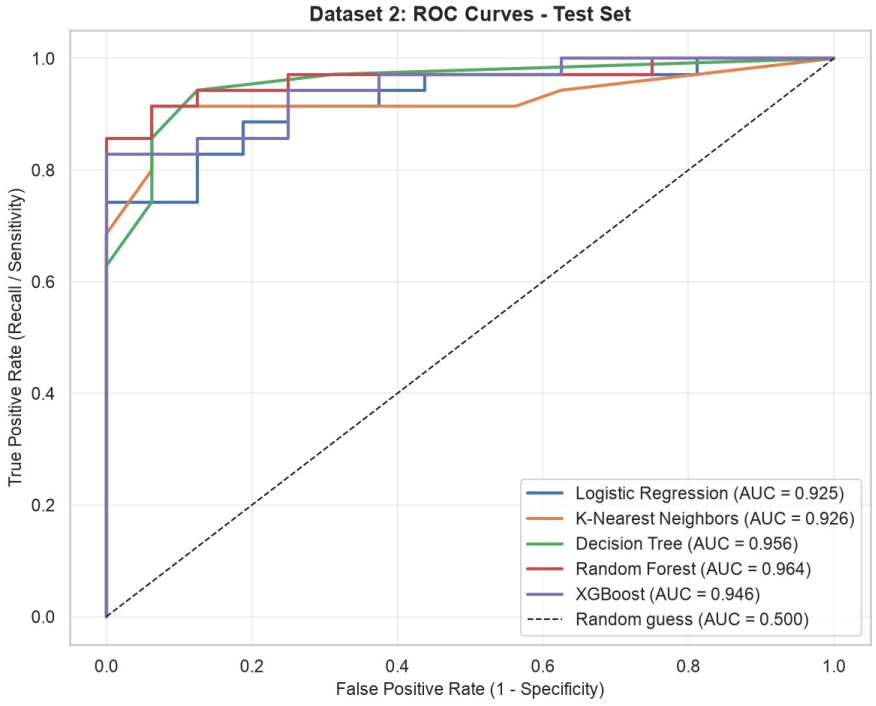


Figure 4.11: ROC curves of the secondary dataset (test set).

All models reach a ROC-AUC above 0.92. Random Forest misses only 3 of the 35 diabetic records with 2 false alarms; XGBoost and KNN raise no false alarms but miss 6 and 11 diabetic records.

Model	Test mean (5 metrics)	CV mean (5 metrics)	Test Recall	CV Recall	Overall score	Recall ≥ 0.70 in both
Random Forest	**0.9298**	**0.9480**	**0.9143**	**0.9421**	**0.9389**	**Yes**
XGBoost	0.9127	0.9550	0.8286	0.9566	0.9338	Yes
Decision Tree	0.9145	0.9076	0.8571	0.9127	0.9111	Yes
Logistic Regression	0.8687	0.9244	0.8000	0.9130	0.8966	Yes
K-Nearest Neighbors	0.8419	0.9016	0.6857	0.8111	0.8717	No

With the **same selection rule** as for Pima, **Random Forest** is selected again. Final model performance (test set): Accuracy 0.9020 · Precision 0.9412 · Recall 0.9143 · F1 0.9275 · ROC-AUC 0.9643 · Specificity 0.875. Cross-validated ROC-AUC: 0.977 ± 0.013. XGBoost is a close second (best cross-validated recall and ROC-AUC), and KNN fails the recall rule. With 51 test records, one record changes recall by almost 3 percentage points.

4.13 Secondary Dataset: Feature Importance and SHAP

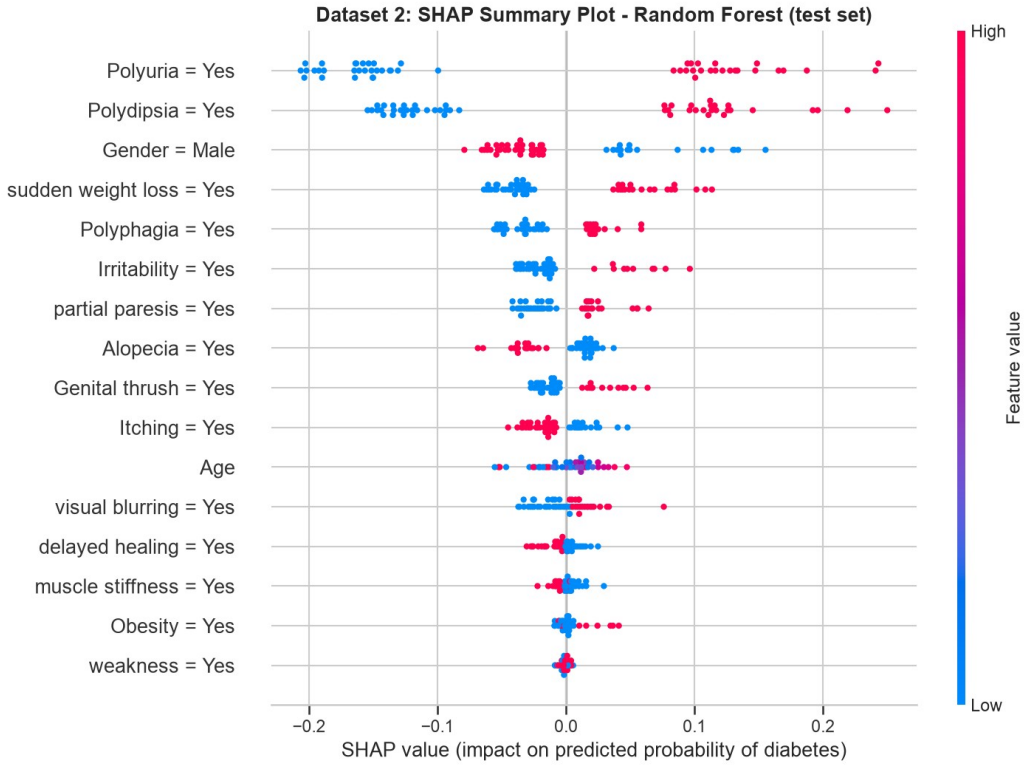


Figure 4.12: SHAP summary plot for the secondary-dataset Random Forest on its test set. Colour = feature value (red = Yes / Male / older).

Feature	Random Forest importance	Mean |SHAP|	"Yes" / higher value pushes prediction
Polyuria = Yes	0.247	0.149	toward diabetes

Feature	Random Forest importance	Mean |SHAP|	"Yes" / higher value pushes prediction
Polydipsia = Yes	0.207	0.123	toward diabetes
Gender = Male	0.066	0.052	toward no diabetes
sudden weight loss = Yes	0.070	0.051	toward diabetes
Polyphagia = Yes	0.042	0.032	toward diabetes
Irritability = Yes	0.043	0.029	toward diabetes
Alopecia = Yes	0.034	0.023	toward no diabetes
Age	0.085	0.017	weak

- **Polyuria** and **Polydipsia** dominate in every model and in SHAP; in the Random Forest they account for almost half of the total importance.
- Male gender and Alopecia push predictions toward *no* diabetes, matching the EDA.
- Age ranks third in impurity importance but has a small SHAP value: impurity importance is known to favour continuous features, and Age is the only one here.
- As for Pima, SHAP explains how the model uses the features; it does not prove medical causality.

4.14 Comparison of the Two Datasets

The same five models, metrics and selection rule were applied to both datasets (results/dataset_comparison.csv). Because the populations and features differ, the comparison describes model behaviour and is not a ranking of the datasets.

Dataset – Model	Accuracy	Precision	Recall	F1	ROC-AUC	CV ROC-AUC
Pima – XGBoost	0.7532	0.6250	0.7407	0.6780	0.8263	0.8217
Pima – Logistic Regression	0.7078	0.5738	0.6481	0.6087	0.8113	0.8427
Pima – Random Forest (final)	**0.7468**	**0.6154**	**0.7407**	**0.6723**	**0.8094**	**0.8290**
Pima – K-Nearest Neighbors	0.6948	0.5455	0.7778	0.6412	0.7974	0.8196
Pima – Decision Tree	0.7273	0.5833	0.7778	0.6667	0.7616	0.7617
Early Stage – Random Forest (final)	**0.9020**	**0.9412**	**0.9143**	**0.9275**	**0.9643**	**0.9772**
Early Stage – Decision Tree	0.8824	0.9677	0.8571	0.9091	0.9562	0.9004
Early Stage – XGBoost	0.8824	1.0000	0.8286	0.9062	0.9464	0.9802
Early Stage – K-Nearest Neighbors	0.7843	1.0000	0.6857	0.8136	0.9259	0.9657
Early Stage – Logistic Regression	0.8235	0.9333	0.8000	0.8615	0.9250	0.9721

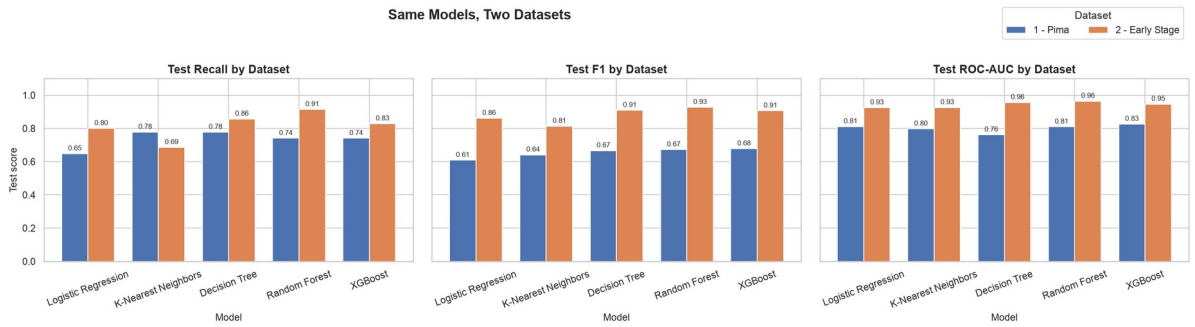


Figure 4.13: Test recall, F1 and ROC-AUC of the same models on both datasets.

- **The secondary dataset is easier to predict:** averaged over the five models, test ROC-AUC is 0.944 against 0.801 for Pima.
- **Important features differ but tell a consistent story:** Glucose dominates Pima; Polyuria and Polydipsia dominate the Early Stage dataset. In both, the strongest predictors are those most directly linked to high blood sugar.
- **The models behave consistently:** the same rule selects Random Forest on both datasets. In cross-validated ROC-AUC the ensembles and Logistic Regression lead and the single Decision Tree is last in both. KNN is the least consistent model (highest test recall on Pima, lowest on the Early Stage dataset).
- **Why performance differs:** symptoms that often accompany established diabetes vs. clinical measurements that overlap between classes; a diabetes-hospital sample (69 % diabetic) vs. a population cohort (35 %); many missing laboratory values in Pima; and a much smaller secondary test set (51 records).
- **Limits of the comparison:** different countries, populations, periods, collection methods and features. A higher score on the secondary dataset does not mean symptom questionnaires are better screening tools in general, and the models cannot be applied to each other's data.

Effect of the duplicate records (sensitivity check). Repeating the secondary-dataset experiment with the duplicates left in places an identical copy of 72 of the 104 test rows (69 %) in the training set, and every model appears better than it is:

Model (secondary dataset)	Accuracy (unique, used)	Accuracy (with duplicates)	ROC-AUC (unique)	ROC-AUC (with duplicates)
Random Forest	0.902	0.971	0.964	0.998
Decision Tree	0.882	0.962	0.956	0.988
XGBoost	0.882	0.971	0.946	1.000
K-Nearest Neighbors	0.784	0.904	0.926	0.995
Logistic Regression	0.824	0.942	0.925	0.991

4.15 Discussion

- **Glucose dominates.** Every analysis (EDA, correlation, coefficients, impurity importance and SHAP) identifies plasma glucose as the strongest signal. This agrees with the clinical role of glucose testing and with prior studies [8], [10].
- **Model families perform similarly.** Test ROC-AUCs of 0.81–0.83 for Logistic Regression, Random Forest and XGBoost suggest that the signal is largely monotonic and can be captured by a simple model. The ensembles gave better recall/precision trade-offs at the default threshold.
- **Imbalance handling matters for recall.** SMOTE raised Logistic Regression's cross-validated recall from 0.575 to 0.724 with little change in ROC-AUC.
- **CV and test can disagree.** Logistic Regression ranked first in cross-validated ROC-AUC but third on the test set, while XGBoost showed the opposite pattern. With 154 test patients, one patient changes recall by about 1.9 points, so selection based on both evaluations is more reliable than relying on the test split alone.
- **Comparison with literature.** The final model's accuracy (0.747) and AUC (0.809) are in the range of simple baselines reported on this dataset (e.g. about 76 % in [2], [7], and a random-forest accuracy of 0.7604 on complete Pima records in [8]). They are below heavily engineered pipelines [9], which use additional steps (outlier removal, feature selection, ensembling) and different validation schemes. Because protocols differ, these numbers are not directly comparable.
- **Two datasets, one method.** The same leak-free methodology worked on both datasets, but each needed its own data-quality step (invalid zeros vs. duplicates) and its own imbalance decision (SMOTE vs. no re-weighting). Random Forest was the most consistent model across both.

Chapter 5 — Conclusion

5.1 Summary

This project built complete, leak-free predictive analytics pipelines for diabetes prediction on two independent datasets, following the KDD process from problem definition to knowledge discovery. On the primary Pima dataset, invalid zeros were treated as missing and imputation, scaling and SMOTE were learned from training data only. On the secondary Early Stage dataset, the target was detected automatically, duplicate records were removed before splitting, and imputation, scaling and one-hot encoding were learned from training data only. The datasets were not merged; five classifiers were compared on each with cross-validation and a hold-out test set.

5.2 Main Findings

- All five models were far better than random guessing on both datasets (test ROC-AUC 0.762–0.826 on Pima, 0.925–0.964 on the Early Stage dataset).
- Handling class imbalance substantially improved recall at a small cost in accuracy.
- Simple and complex models performed similarly in ranking ability. The ensembles gave the best balance of recall and precision at the default threshold.
- The symptom-based secondary dataset was clearly easier to predict (average test ROC-AUC 0.944 vs 0.801), and removing its duplicate records prevented scores from being inflated to about 97 % accuracy.

5.3 Best Model

Random Forest was selected for both datasets with the same recall-aware, multi-metric rule. **Pima:** accuracy 0.747, precision 0.615, recall 0.741, F1 0.672, ROC-AUC 0.809 (40 of 54 diabetic patients caught; cross-validated ROC-AUC 0.829 ± 0.025). **Early Stage:** accuracy 0.902, precision 0.941, recall 0.914, F1 0.928, ROC-AUC 0.964 (32 of 35 diabetic records caught; cross-validated ROC-AUC 0.977 ± 0.013).

5.4 Important Features

Pima: Glucose is the dominant predictor, followed by BMI, Age and the Diabetes Pedigree Function. **Early Stage:** Polyuria and Polydipsia dominate, followed by gender and sudden weight loss. In both, the strongest predictors are those most directly linked to high blood sugar. These are predictive associations, not causal effects.

5.5 Limitations

- Small dataset (768 patients; 154 in the test set), so metrics carry substantial uncertainty.
- Narrow demographics (Pima Indian women aged 21 or older) and historical data; generalisation to other populations is unknown.

- Possible sampling bias from a single population study.
- Heavy missingness in Insulin (48.7 %) and SkinThickness (29.6 %), filled by median imputation.
- SMOTE samples are synthetic, not real patients.
- No exhaustive tuning, a fixed 0.5 threshold, uncalibrated probabilities and no external validation.
- Secondary dataset: only 251 unique records (51 in the test set), a hospital-based sample with self-reported symptoms, and duplicates whose origin is unknown.
- The two datasets differ in population, country, period, collection method and features, so they can be compared only at the level of model behaviour.
- The model is an academic screening demonstration, not a medical diagnostic tool.

5.6 Future Scope

Larger and more diverse datasets; external validation; cross-validated or nested hyperparameter optimisation; per-patient explanations (SHAP waterfall plots); probability calibration and cost-sensitive threshold selection; additional clinical features such as HbA1c; deployment as a screening-support tool with appropriate oversight; and monitoring for data and model drift after deployment; a larger symptom dataset without duplicates; and a dataset that records both clinical measurements and symptoms for the same people.

5.7 KDD Mapping

KDD Stage	Primary dataset (Pima)	Secondary dataset (Early Stage)
Problem Definition	Diabetes onset (Outcome 0/1) from clinical measurements	Diabetes (class Positive/Negative) from symptoms
Data Selection	Pima Indians Diabetes Dataset (768 × 9)	UCI Early Stage Diabetes Risk Prediction (520 × 17)
Data Cleaning	Invalid zeros → NaN	Text clean-up, 269 duplicates removed (251 unique)
Data Transformation	Split, median imputation, scaling, SMOTE (train only)	Split, imputation, scaling, one-hot encoding (train only)
EDA	Distributions, by-outcome plots, correlations	Symptom prevalence, gender, age, associations
Modeling	LR, KNN, DT, RF, XGBoost	LR, KNN, DT, RF, XGBoost
Evaluation	Accuracy, Precision, Recall, F1, ROC-AUC, CV	Same metrics and selection rule
Interpretation	Feature importance + SHAP	Feature importance + SHAP
Knowledge Discovery	Glucose dominates; Random Forest selected	Polyuria and Polydipsia dominate; Random Forest selected

Chapter 6 — References

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